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Symmetric Encryption

Overview

Symmetric encryption uses one secret key both to lock and to unlock the data. It is fast and suits large amounts of data such as files and disks. The widely used algorithm is AES. In Python the cryptography package offers Fernet, a safe, easy recipe: key = Fernet.generate_key(), then Fernet(key).encrypt(data) and decrypt(token). The difficulty is the key: anyone who has it can read the data, so it must be stored and shared securely, never written into the source code.

Key points

- Part of: Encryption in Python
- About 25 minutes
- Try the example yourself, then check it

How to use this handout

1. Read the overview first, then work through the key points in order.
2. Write down anything you are unsure about and ask in the course forum.
3. Use page 2 to plan your practice and keep notes.

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Practice checklist

- [ ] Read: Symmetric Encryption
- [ ] Try the example from the lesson
- [ ] Explain the main idea in your own words
- [ ] Answer the related questions
- [ ] Ask about anything unclear in the forum

My notes